Quantum-Enhanced Strategic Siting of Energy Storage and Microgrids

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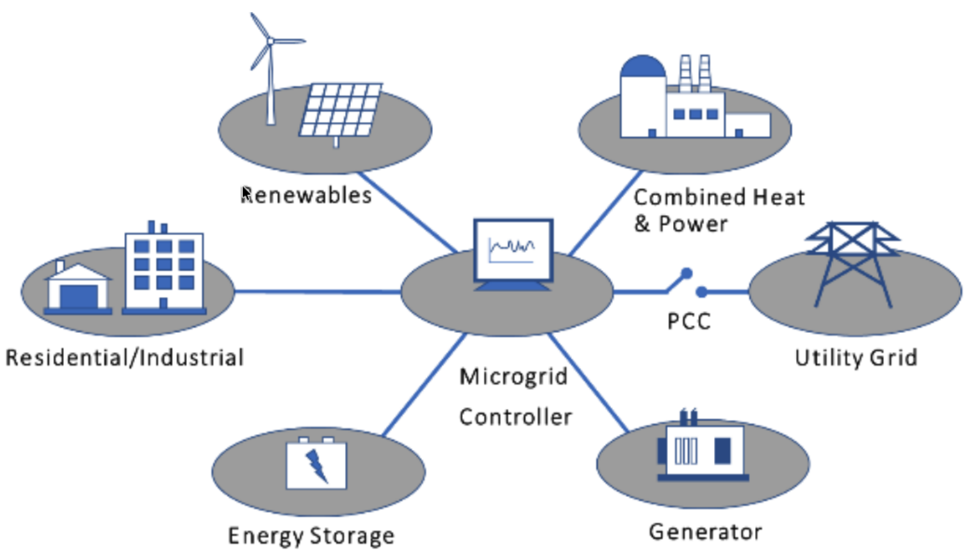

Figure 1: Microgrid architecture and components. Source: S. Atcitty, U.S. Department of Energy presentation.

Team Name

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Strategic Context
This Phase 1 concept targets the fundamental computational bottleneck in grid modernization: solving large-scale mixed-integer nonlinear programming (MINLP) instances that determine where energy storage and microgrids should be deployed, how large they should be, and how they should operate under uncertainty. The candidate portfolio must satisfy nonlinear AC power-flow physics, contingency constraints, and multi-scenario cost and resilience trade-offs. We therefore frame the problem as a quantum-accelerated planning workflow that can explore feasible infrastructure portfolios faster than classical enumeration while preserving utility-grade validation.

Proposed Solution
The proposed solution maps the discrete planning layer into Quadratic Unconstrained Binary Optimization (QUBO) and Ising Hamiltonian formulations so that binary siting, sizing, and topology decisions become native inputs to quantum search. Continuous dispatch variables, AC feasibility, and reliability constraints remain on classical solvers. This separation keeps the hardest combinatorial search in the quantum loop while preserving full-fidelity power-system validation in the classical loop. Our deliverable will be a working prototype that returns ranked siting and sizing portfolios, scenario-aware operating summaries, AC-feasibility certificates, and Pareto trade-off reports.

Hybrid Optimization Workflow
The iteration follows a Hybrid Quantum-Classical Benders Decomposition (HQCBD) workflow. Rather than a single-objective QUBO only, we use a Quantum-Assisted Multi-Objective Optimization (QAMOO) master in which weighted graph objectives are sampled and refined across scenarios:

$$
\begin{aligned}
\max_{x,e} \quad \lambda^\top \mathbf{f}(x) &= \sum_{k=1}^{m} \lambda_k f_k(x), \qquad \sum_{k=1}^{m} \lambda_k = 1,\ \lambda_k \geq 0, \\
f_k(x) &= \sum_{(i,j)\in E_k} w_{ij}^{(k)} e_{ij}, \quad x_i \in \{0,1\},\ e_{ij} \in \{0,1\},
\end{aligned}
$$

The binary edge variables are linked to the partition decisions through standard linear constraints. For each scenario, classical feasibility checks enforce AC limits and operating bounds, while the quantum loop explores candidate portfolios and refines the Pareto set through scenario-by-scenario sampling. Those candidates are then translated into the Ising representation used by the quantum optimizer, and infeasible or weak solutions produce feasibility and optimality cuts that tighten the next master iteration.

Technical Stack and Implementation Path
Qiskit is the quantum programming layer for the prototype. It is used to build problem-specific ansatzes, manage circuit parameters and shot budgets, and connect quantum samples to classical feasibility and optimization steps. The implementation starts on reduced utility test systems and scales by increasing candidate sites, scenario count, and cut complexity.

The prototype will be validated against classical baselines on the same reduced instances, with the goal of showing better candidate quality and faster iteration on the same planning problems[1].

Conclusion
The proposed approach positions quantum methods as a targeted accelerator inside an existing planning pipeline, not as a replacement for grid-physics modeling. By combining Qiskit-based quantum search with rigorous classical feasibility analysis and multi-objective scenario aggregation, the team can move from concept to code quickly and establish a credible path to larger, decision-relevant grid-planning studies.

[1]Primary reference: Nature Machine Intelligence article, https://www.nature.com/articles/s43588-025-00873-y.